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Utilization and State Tracking Application

Version 1.0.5

Abstract

This example provides (in Building Block design pattern) both the infrastructure (shapes, data entities and services) and application for a generic state tracking utilization solution that can be used as a demonstration and / or for a proof of concept / pilot. Three implementations are included to accommodate storing data in either ThingWorx data objects (data tables and streams), PostgreSQL database objects, and SQL Server data objects.

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Best Practice Intended Use

This example provides (in Building Block design pattern) both the infrastructure (shapes, data entities and services) and application for a generic state tracking utilization solution that can be used as a demonstration and / or for a proof of concept / pilot. Three implementations are included to accommodate storing data in either ThingWorx data objects (data tables and streams), PostgreSQL database objects, and SQL Server data objects.

Use Case

The use case is simple machine state tracking along with a utilization calculation for a selected time frame. This use case is purposely less complex for the situation where state tracking is the desired use case and Utilization Percent is the desired KPI. For use cases that are required to track OEE along with different shifts and products, PTC has existing OOTB solutions to address these use cases:

- Real Time Production Performance Monitoring (RTPPM)
- Digital Performance Management (DPM)

Intended Persona/Process

Customers who want to do simple state tracking but are not required to calculate OEE at this point.

Intended Business Value

This application is provided as an example of how to do state tracking when OEE is not desired. The application will need to be extended when enough machines are tracked to warrant a hierarchical approach to visualization. An alternative approach would be to use the Utilization and State Tracking Application within the Smart Connected Product Accelerator (SCPA) framework to leverage its asset network and organizational capabilities. SCPA has an additional UI building block for the Utilization and State Tracking Application to leverage the SCPA asset network. This document will provide information for SCPA and non-SCPA deployments.

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Prerequisites and Caveats

The following lists recommended experience for using the application:

- Deployment of the Application
 - ThingWorx Development Experience
 - Setting Permissions in the Solution Framework
 - Basic Postgres or SQL Server skills required if using those application implementations
 - Familiarity with SCPA if deployed within a SCPA environment
- Customization of the Application
 - ThingWorx Development Experience
 - Knowledge of Building Blocks

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Overview of Best Practice

Supplied Files

1. These are the Building Block extension files:
 - a. utilization-thingworxobjects-1.0.5-extension-package.zip
 - b. utilization-sqlobjects-1.0.5-extension-package.zip
 - c. utilization-postgresqlobjects-1.0.5-extension-package.zip
 - d. SCPA-utilization-postgressqlobjects-1.0.5-extension-package.zip
 - e. SCPA-utilization-sqlobjects-1.0.5-extension-package.zip
 - f. SCPA-utilization-thingworxobjects-1.0.5-extension-package.zip
2. UtilizationBuildingBlockApplication.1.0.5.pptx
 - a. PowerPoint that runs through the provided functionality
3. UtilizationApplicationUserManual.1.0.5.pdf
 - a. This document
4. SQLTableAndProcedureScripts.sql
 - a. Scripts for creating the SQL Server data infrastructure
5. PostgreSQLScripts.sql
 - a. Scripts for creating the PostgreSQL data infrastructure
6. Permissions Entities (for non-SCPA deployments):
 - a. PTCSC.UtilizationTWImpl.SFPermissions.xml
 - b. PTCSC.UtilizationSQLImpl.SFPermissions.xml
 - c. PTCSC.UtilizationPostgresImpl.SFPermissions.xml
7. UtilizationBuildingBlockAppPermissions.xlsx
 - a. Utilization Building Block Application Permissions Reference

Building Block Design Pattern

1. The following building blocks were created for the application:

Building Block Name	Building Block Type	Description
PTCSC.Utilization	Abstract	Services building block
PTCSC.UtilizationTWImpl	Implementation	Implementation of PTCSC.Utilization using ThingWorx data objects
PTCSC.UtilizationSQLImpl	Implementation	Implementation of PTCSC.Utilization using SQL Server data objects
PTCSC.UtilizationPostgresImpl	Implementation	Implementation of PTCSC.Utilization using PostgreSQL data objects
PTCSC.UtilizationUI	UI	Utilization UI components

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PTCSC.UtilizationAdminUI	UI	Building block to drive the Utilization Admin menu options
PTCSC.UtilizationUIForSCPA	UI	Additional UI building block with SCPA specific content (SCPA deployments only)

2. PTCSC.Utilization Details

- This is the abstract building block
- Data Shapes are defined in this building block
- Services are defined in the PTCSC.Utilization.Management_TS of this building block
- A PTCSC.Utilization.ModelLogic_TS is defined to be applied to model entities in the solution

3. PTCSC.UtilizationTWImpl Details

- This is the implementation building block that uses ThingWorx data objects
- Service code lives in the PTCSC.UtilizationTWImpl.Manager_TT
- ThingWorx Data Objects
 - PTCSC.UtilizationTWImpl.MachineCode_DT
 - PTCSC.UtilizationTWImpl.Reason_DT
 - PTCSC.UtilizationTWImpl.ReasonGroup_DT
 - PTCSC.UtilizationTWImpl.UtilizationState_DT
 - PTCSC.UtilizationTWImpl.Utilization_SM

4. PTCSC.UtilizationSQLImpl Details

- This is the implementation building block that uses SQL Server data objects
- Service code lives in the PTCSC.UtilizationSQLImpl.Manager_TT
- The services call into the PTCSC.UtilizationSQLImpl.SQLDatabase thing
- The PTCSC.UtilizationSQLImpl.Database thing has its services defined on the PTCSC.UtilizationSQLImpl.SQLDatabase_TT which is derived from a SQLThing
 - Since the PTCSC.UtilizationSQLImpl.Database thing is derived from a SQLThing, we need to tie it to an existing persistence provider. By using an existing persistence provider, we can cut down on the number of database connections.

5. PTCSC.UtilizationPostgresImpl Details

- This is the implementation building block that uses PostgreSQL data objects
- Service code lives in the PTCSC.UtilizationPostgresImpl.Manager_TT
- The services call into the PTCSC.UtilizationPostgresImpl.SQLDatabase thing
- The PTCSC.UtilizationPostgresImpl.SQLDatabase thing has its services defined on the PTCSC.UtilizationPostgresImpl.SQLDatabase_TT which is derived from a SQLThing
 - Since the PTCSC.UtilizationPostgresImpl.SQLDatabase thing is derived from a SQLThing, we need to tie it to an existing persistence provider. By using an existing persistence provider, we can cut down on the number of database connections.

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Upgrade Steps

1. If upgrading from a previous version of the Utilization Application, simply import the new extension file that matches the previous implementation that was installed. There was one small change to the database scripts to change the Failure column of the Reason table to allow nulls. This avoids an error on reason record add when the Failure value is null (neither true nor false). The SQL Server script for the AddOrUpdateUtilizationDataRecord stored procedure was changed to a MERGE statement with concurrency locks.
2. Deploying the upgraded Utilization Building Blocks
 - a. Navigate to the PTC.Base.Manager thing
 - b. Go to services
 - c. Run the InitializeSolution service with an empty JSON object as the deploymentConfig input {}
3. Configure user permissions appropriately in the Solution Framework (if not using SCPA). If using SCPA, permissions should be set appropriately already.
 - a. Import Utilization application permissions (if not using SCPA)
 - i. In Composer, import the appropriate provided permissions file for the implementation you are using:
 1. ThingWorx Data Objects: PTCSC.UtilizationTWImpl.SFPermissions.xml
 2. SQL Data Objects: PTCSC.UtilizationSQLImpl.SFPermissions.xml
 3. Postgres Data Objects: PTCSC.UtilizationPostgresImpl.SFPermissions.xml
 - b. If you would prefer to set the permissions manually in Solutions Framework, reference the provided UtilizationBuildingBlockAppPermissions.xlsx file.
4. Starting with Utilization 1.0.4 there is Utilization Overview mashup with a filtering capability. If upgrading, the original mashup will be associated with the Utilization Overview menu option. To switch to the new mashup with filtering capability:
 - a. Go to Solution Framework
 - b. Scroll to the PTCSC.UtilizationUI building block and select the row
 - c. Click the View button
 - d. On the left, select the Single Pane option under Configure
 - e. Scroll to the Menu Configuration section
 - f. Select the Utilization Overview row
 - g. Click the Edit Menu Item button
 - h. In the pop-up:
 - i. Select Utilization Overview as the Menu Title
 - ii. Select PTCSC.UtilizationUI.MainWithSelection_MU as the Menu Mashup Name
 - iii. Click the Save button
 - i. Refresh your browser and the new screen should be selected
 - j. The original mashup is still available if preferred (PTCSC.UtilizationUI.Main_MU)

New Installation Steps

1. For new installations, determine which implementation you will utilize
 - a. PTCSC.UtilizationTWImpl – configuration data stored in 4 ThingWorx data tables and historical data stored in a ThingWorx stream object (not a value stream, but a stream with schema)

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- b. PTCSC.UtilizationSQLImpl – configuration and historical data stored in SQL Server
- c. PTCSC.UtilizationPostgresImpl – configuration and historical data stored in PostgreSQL
- 2. If installing to an existing SCPA framework, the required base building blocks and generic permissions components will already be installed.
- 3. If not using SCPA, install and initialize Solution Framework 1.2.102 (or later)
 - a. [Link for Solution Framework 1.2.102](#)
 - b. Install instructions are included in the above link for the Solution Framework. We simply:
 - i. Downloaded the solutionframework-1.2.102-extension-package.zip
 - ii. Imported the file into our ThingWorx server as an extension
 - iii. Ran the InitializeSolution service on PTC.Base.Manager with the proper JSON input
 - iv. This was the JSON for a SQL Server ThingWorx installation:

```
{
  "twxAdminUserName" : "Administrator",
  "twxAdminPassword" : "passwordhere",
  "twxURL" : "https://serveraddress/Thingworx",
  "databaseUser": "twadmin",
  "overrideComponentDeploymentState": false,
  "databasePassword": "twadmin",
  "databaseJDBCString":
  "jdbc:sqlserver://localhost:1433;databaseName=thingworx",
  "databaseThing": "PTC.DBConnection.SQLThingDatabase"
}
```

- v. This was the JSON for a Postgres ThingWorx installation:

```
{
  "twxAdminUserName" : "Administrator",
  "twxAdminPassword" : "passwordhere",
  "twxURL" : "https://serveraddress/Thingworx",
  "databaseUser": "twadmin",
  "overrideComponentDeploymentState": false,
  "databasePassword": "twadmin",
  "databaseJDBCString": "jdbc:postgresql://localhost:5432/thingworx",
  "databaseThing": "PTC.DBConnection.PostgresDatabase"
}
```

- 4. Import the desired implementation extension package file in Composer

Framework	Extension Package Name	Data Structure Type
Non-SCPA	utilization-thingworxobjects-1.0.5-extension-package.zip	ThingWorx data objects
Non-SCPA	utilization-sqlobjects-1.0.5-extension-package.zip	SQL server data objects
Non-SCPA	utilization-postgresqlobjects-1.0.5-extension-package.zip	PostgreSQL data objects

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SCPA	SCPA-utilization-thingworxobjects-1.0.5-extension-package.zip	ThingWorx data objects
SCPA	SCPA-utilization-sqlobjects-1.0.5-extension-package.zip	SQL server data objects
SCPA	SCAP-utilization-postgresqlobjects-1.0.5-extension-package.zip	PostgreSQL data objects

5. If using SQL Server data objects:

- a. Go to your SQL Server
- b. Create a database to house your Utilization objects
- c. Run the provided SQLTableAndProcedureScripts.sql against that database to create the required database tables and stored procedures
- d. In Composer, make sure that you have a persistence provider configured to your SQL database (make sure that you set it to “active” as well)
 - i. If you create a new persistence provider, make sure to check the configuration settings. These were the settings on a test Persistence Provider:

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General Information
Services
Configuration
Permissions

Configuration

Connection Information

JDBC URL

Username

Password

Initial Connection Pool Size

Connection Acquire Increment

Max Connection Pool Size

Min Connection Pool Size

Max Cached Statements

Driver Class

Acquire Retry Attempts

Acquire Retry Delay

Checkout Retry Timeout

Pooled Connection Idle Re-Test Period (seconds)

☐ Test Connections On Pool Checkout

- ii. After configuration of the persistence provider, if you run the TestConnectivity service you should get true:

Output		+ Data Shape
result (1)		
connectionData	status	
jdbc:sqlserver://EC2AMAZ-H2QIN06\MSSQL:1433;databaseName=Utilization;applicationName=Thingworx;	☒	

- e. Go to the PTCSC.UtilizationSQLImpl.SQLDatabase thing
- f. Go to Configuration, choose the appropriate persistence provider that connects to your SQL database. Save. Go to services and run the RestartThing service.

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- g. Once you set and save this configuration, if you run the GetUtilizationStates service on the PTCSC.UtilizationSQLImpl.SQLDatabase thing, you should get zero rows, but it should connect and list the columns:

Output + Data Shape			
result			
UtilizationState	UtilizationStateDescription	IncludeInUtilizationCalculation	Utilized
No data			

6. If using PostgreSQL data objects
- Go to your PGAdmin interface
 - Create a database to house your Utilization objects
 - Run the provided PostgreSQLScripts.sql against that database to create the required database schema and tables
 - In Composer, make sure that you have a persistence provider configured to your database
 - If you create a new persistence provider, make sure to check the configuration settings (make sure that you set it to “active” as well). These were the settings on a test Persistence Provider:

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General Information
Services
Configuration
Permissions

Configuration

Connection Information

JDBC URL
jdbc:postgresql://localhost:5432/utilization

Username
postgres

Password

Initial Connection Pool Size
5

Connection Acquire Increment
5

Max Connection Pool Size
100

Min Connection Pool Size
5

Max Cached Statements
100

Driver Class
org.postgresql.Driver

Acquire Retry Attempts
3

Acquire Retry Delay
10000

Checkout Retry Timeout
1000000

Pooled Connection Idle Re-Test Period (seconds)
60

- ii. After configuration of the persistence provider, if you run the TestConnectivity service you should get true:

Output		+ Data Shape
result (1)		
connectionData	status	
jdbc:postgresql://localhost:5432/utilization?sslmode=disable	☒	

- e. Go to the PTCSC.UtilizationPostgresImpl.SQLDatabase thing

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- f. Go to Configuration, click edit, and choose the appropriate persistence provider that connects to your PostgreSQL database. Save. Go to services and run the RestartThing service.
- g. Once you set and save this configuration, if you run the GetUtilizationStates service on the PTCSC.UtilizationPostgresImpl.SQLDatabase thing, you should get zero rows, but it should connect and list the columns:

Output + Data Shape			
result			
Utilization State	Utilization StateDescription	IncludeInUtilizationCalculation	Utilized
No data			

7. Deploying the Utilization Building Blocks
 - a. Navigate to the PTC.Base.Manager thing
 - b. Go to services
 - c. Run the InitializeSolution service with an empty JSON object as the deploymentConfig input {}
8. Configure the Specific Utilization Manager
 - a. Navigate to the specific implementation manager that was chosen (PTCSC.UtilizationSQLImpl.Manager, PTCSC.UtilizationPostgresImpl.Manager or PTCSC.UtilizationTWImpl.Manager)
 - b. Go the configuration tab
 - c. Enter or edit configuration values as needed:
 - i. DaysToKeepWithPurging – integer (7 is the default)
 - ii. HoursToLookBackOnUtilScreens – integer (8 is the default)
 - iii. InitialScreenRefreshDisabled – boolean (true is the default)
 - iv. ScreenRefreshRate – integer (30 seconds is the default)
 - v. UtilizationCriticalPercent – integer value at which formatted utilization value goes from red to yellow (50 is the default)
 - vi. UtilizationWarningPercent – integer value at which formatted utilization value goes from yellow to green (63 is the default)
 - vii. MaxEntitiesOnOverviewScreen – integer value to limit the number of individually listed machines on the Overview Screen (10 is the default)
 - viii. DemoMode – boolean (true is the default). If true, the machine code admin screen will have the option “use simulated data for a machine”. Also, if true the simulation logic will not be skipped.
 - ix. ShiftFiltersEnabled – boolean (false is the default). If true, shift filters will be present on the application screens
 - x. ReasonCodeEditEnabled – boolean (false is the default). If true, all utilization reasons can be edited.
 - xi. UseReasonCodeChangeSubscription – boolean (true is the default). If true, the subscription on PTCSCReason property value change will create a Utilization Event record.

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- xii. FilterReasonCodeEditByState – boolean (false is the default). If true, when editing a utilization record, only reasons that have the same state as the current reason will be shown and available for editing.
- xiii. ShiftConfiguration
 - 1. This is a configuration table that houses the ShiftID and the Shift Description that will be used for Shift Filters (if enabled). The ShiftID will be what is contained in the utilization record. The ShiftDescription will be displayed in the dropdown. The value “All Shifts” should never be used for the ShiftID as this will automatically be at the top of the filter list and will show all values.
- xiv. UtilizationStateColorConfiguration
 - 1. The utilization schedule (Gantt) charts can be passed in specific colors. In an effort to make the application screen colors more consistent, colors were configured to match the utilization state bar chart, pie chart, and grids colors. These will be loaded by default. If you wish to return to the colors in prior version of the building block, you can configure the colors as appropriate.
 - 2. New version utilization schedule (Gantt) chart colors (v1.0.5 and later):
 - a. Down – #ee8686
 - b. Idle – #f3cb78
 - c. Running – #9ed287
 - d. Setup – #53acff
 - e. Unavailable – #adb5bd
 - 3. Earlier version utilization schedule (Gantt) chart colors (v1.0.4 and earlier):
 - a. Down – Red
 - b. Idle – Yellow
 - c. Running – MediumSeaGreen
 - d. Setup – LightSkyBlue
 - e. Unavailable – Silver
- d. Save the changes if edited
- 9. Configure user permissions appropriately
 - a. If using Solution Framework (not using SCPA)
 - i. Import Utilization application permissions (if not using SCPA)
 - 1. In Composer, import the appropriate provided permissions file for the implementation you are using:
 - a. ThingWorx Data Objects: PTCSC.UtilizationTWImpl.SFPermissions.xml
 - b. SQL Data Objects: PTCSC.UtilizationSQLImpl.SFPermissions.xml
 - c. Postgres Data Objects: PTCSC.UtilizationPostgresImpl.SFPermissions.xml
 - 2. If you would prefer to set the permissions manually in Solutions Framework, reference the provided UtilizationBuildingBlockAppPermissions.xlsx file.
 - ii. In Solution Framework (the PTC.SolutionFramework.Main_MU mashup), click on the Global View button (upper right)
 - iii. In the left menu, click Role Management
 - iv. Add appropriate users to the ApplicationAdmin role as well as the AllUsers role

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1. AllUsers role will be able to see the Utilization screens
2. ApplicationAdmin role will be able to see the Utilization screens and the Admin Configuration screens
- v. For testing, we setup a utiladmin user in the ApplicationAdmin role and a utiluser in the AllUsers role
- b. If using SCPA
 - i. The SCPA project files provided contain the proper permissions

Setup for Demo

1. We will need a template to create model objects from that we want to collect utilization data from
2. Simply create a new template (i.e. PTCSC.Utilization.Asset_TT)
 - a. Create from an appropriate base template like GenericThing or RemoteThing.
3. For the created template, give Visibility permissions for the template and instances to:
 - a. PTCCTS.BasePermissions.Default_OR (if not using SCPA)
 - b. If using SCPA, utilize model objects that have already been configured in SCPA
4. Add the PTCSC.Utilization.ModelLogic_TS thing shape to templates that you would like to provide utilization capability to
5. Create some instances from the template (i.e. PTCSC.Utilization.Lathe1, PTCSC.Utilization.Lathe2, etc.)
6. The description will be the name displayed on the mashups
7. For each thing that is created, go to the PTCSCSimulationEnabled property and set it to true (or use the check box on the Machine code admin screen.
8. Additional Machine code configuration steps
 - a. An integer machine code can be used to drive a lookup to determine Utilization State, Reason Group and Reason
 - b. If you would like to add this to your demo data, complete the following:
 - i. On the PTCSC.UtilizationTWImpl.Manager, PTCSC.UtilizationSQLImpl.Manager, or PTCSC.UtilizationPostgresImpl.Manager thing, run the AddDemoMachineCodeDataForEquipment service for each piece of equipment (use the thingname as the EquipmentID parameter) you would like to use MachineCode to drive the state lookup
 - ii. Set the property PTCSCUseMachineCode on the thing to true (or use the Machine Code Admin mashup to set this property)
9. Review the provided Power Point file for functionality overview
10. Additional Machine(s)
 - a. If additional machine(s) are desired, simply create another / more instance(s) from the Thing Template that was created above and configure appropriately. Nothing else is required.
11. Data Simulation
 - a. The subscription on the PTCSC.Utilization.ModelLogic_TS thing shape to the PTCSC.Utilization.Simulation_TM timer event is what runs the utilization data simulation. The timer event is configured to run every minute. If simulation records are not appearing, make sure that the

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entity is configured to use simulation and that the subscription is enabled / timer is enabled. Also make sure that the DemoMode configuration is set to true.

12. Utilization Record Purging

- a. The PTCSC.Utilization.RecordPurge_TM timer thing is set at 86400000 ms (1 day)
- b. The subscription to the timer event is on the implementation manager thing (PTCSC.UtilizationTWImpl.Manager, PTCSC.UtilizationPostgresImpl.Manager or PTCSC.UtilizationSQLImpl.Manager) and will delete utilization records older than the configured threshold (7 days is the default).
- c. If utilization records are not being purged, check to make sure that this subscription and the timer is enabled.

Setup for a Proof of Concept / Pilot

1. Disable the subscription on the PTCSC.Utilization.ModelLogic_TS thing shape to the PTCSC.Utilization.Simulation_TM event.
2. Navigate to the specific implementation manager that was chosen (PTCSC.UtilizationSQLImpl.Manager, PTCSC.UtilizationPostgresImpl.Manager or PTCSC.UtilizationTWImpl.Manager)
 - a. Go the configuration tab
 - b. Edit the DaysToKeepWithPurging parameter if more than / less than 7 days of history is required
 - c. Set DemoMode to false (uncheck)
3. Create a thing template to create your thing instances from
 - a. If you will be connecting to Kepware, this template would use the RemoteThing thing template as a base template
 - b. For the created template, give Visibility permissions for the template and instances to:
 - i. PTCSTS.BasePermissions.Default_OR (or configure the template in SCPA)
 - c. Give it the PTCSC.Utilization.ModelLogic_TS thing shape
4. Determine the required Utilization States, Reason Groups, Reasons required for the POC
5. Make any changes to these items in the Utilization Admin, General Admin administration screen
6. If the Utilization States are altered, several StateDefinitions related to utilization states will need to be adjusted and reimported appropriately on the mashups – not that big of a deal, but hopefully these 5 states are adequate to avoid additional work
 - a. If these changes are required, you will need to duplicate the affected mashups and State Definitions and change your menu configuration.
7. Create instances from your thing template for the machines for the POC
8. If needed, create additional properties on the template or individual things for mapped Kepware tags
9. Install Kepware, and create a corresponding Industrial Gateway Thing in ThingWorx
10. Add the IndustrialThingShape on the thing template that was created
11. Point the instance IndustrialThing property to the Industrial Connector for the Kepware server
12. Determine if a single integer machine code can drive utilization state, reason group and reason. If so, use these steps:
 - a. Map the integer automation tag to the thing property PTCSCMachineCode

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- b. Use the Utilization Admin, Machine Code administration screen to create the lookup table by machine that maps the integer machine code to a state, reason group and reason
- c. Ensure the that “Use Machine Code to Determine Reason Code” check box is checked and the “Set Properties” button is clicked.
- d. The subscription should already be in place and active to look up the required state, reason group and reason when the PTCSCMachineCode value changes. This subscription is on the PTCSC.Utilization.ModelLogic_TS

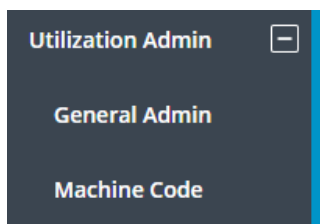
13. If multiple tags drive the state reason:

- a. Map any “tags of interest” that will factor into business logic for change of state / reason code.
- b. Create a UtilizationStateChange service (may be different for specific thing instances) that is triggered when applicable tags change that may signal a utilization state change
- c. The current logic is that the utilization record is recorded when the reason changes.
- d. In the custom service, ensure that the utilization state and reason group properties are set prior to changing the reason
- e. When the reason is changed, the utilization record will be created by the existing subscription / service
- f. Here is an example of a super simple UtilizationStateChange service:

```
if (me["PLC_RUNNING"] == true) {
    me.PTCSCReasonGroup = "Running";
    me.PTCSCUtilizationState = "Running";
    me.PTCSCReason = "Good Production";
}
else if (me["PLC_RUNNING"] == false) {
    me. PTCSCReasonGroup = "Downtime";
    me. PTCSCUtilizationState = "Down";
    me.PTCSCReason = "Unknown";
}
```

Utilization Administration Screens

This section gives an overview of the two Utilization Administration screens. If your Utilization application is not implemented with SCPA, the screens are accessed from the left-hand menu as seen below:



If implemented with SCPA, the screens will be accessed from this section of the SCPA Admin menu:

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Admin

Search

Utilization

General Admin

Machine Code

1. General Admin – this screen allows for the administration of the utilization master data. There are three main objects in the master data: Utilization States, Reason Groups and Reasons.

a. Utilization States – the left side of the screen allows for the add, edit and deletion of utilization states:

Edit States

Utilization State	Utilization State Description	Utilization Calculation Included	Utilized
Down	Machine scheduled or available, but faulted	☒	☐
Idle	Machine scheduled or available, but not running	☒	☐
Running	Machine running	☒	☒
Setup	Machine in changeover	☒	☐
Unavailable	Machine not scheduled or otherwise unavailable	☐	☐

Utilization State:
 Down

Utilization State Description:
 Machine scheduled or available, but faulted

☒ Included in Utilization Calculation

☐ Utilized

Apply
Delete

i. Utilization State – Utilization states are the high-level machine states indicating the general condition of the machine. There are 5 states that are included with the utilization application:

1. Down

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2. Idle
 3. Running
 4. Setup
 5. Unavailable
- ii. Utilization Calculation – the utilization metric is calculated by dividing total time by utilized time
 1. Utilized Time – if a state is flagged “Utilized”, all events with that state will count as utilized time. The Running state is the only state that is flagged as utilized time by default.
 2. Total Time – if a state is flagged “Included in Utilization Calculation”, all events with that state will be included as part of the total time. The Unavailable state is the only state that is flagged false for “Included in Utilization Calculation”.
 - iii. Edit State – you can edit a state by selecting the state record in the grid, making changes and then clicking the Apply button. For an existing state, these are the fields that can be edited:
 1. Utilization State Description
 2. Included in Utilization Calculation
 3. Utilized
 - iv. Add State – Unique states are driven by the name of the state. A new state can be created by entering a new state name in the Utilization State text box and clicking the Apply button.
 - v. Delete State – to delete a state, select the record in the grid and click the Delete button.
 - vi. Important Note on States and Application Color Formatting
 1. Required State
 - a. When there is no utilization data for a machine for a specified time frame, the data will return records with the state of Unavailable. Therefore, this is a required state and should not be deleted.
 2. State Color Formatting
 - a. All widgets on the utilization screens use color formatting for the five utilization states that are supplied with the application:
 - i. Down
 - ii. Idle
 - iii. Running
 - iv. Setup
 - v. Unavailable
 - b. If additional states are created, the application will need to be customized to properly represent new states with appropriate colors.
 - c. The schedule chart widget uses its own color formatting that is described above in the UtilizationStateColorConfiguration section. Every state that is defined needs to have a configured color.
 - b. Reason Groups – the center of the screen allows for the add, edit and deletion of reason groups:

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Edit Reason Groups

Downtime

Planned Downtime

Unplanned Downtime

Machine Fault

Idle

Planned Idle Time

Unplanned Idle Time

Running

Unplanned Running Time

Reason Group:

Downtime

Reason Group Description:

General Group for Downtime

Parent Reason Group:

Apply

Delete

- i. Reason Groups – Reason Groups are simply a hierarchy to organize Reasons
 - ii. Edit Reason Group –you can edit a Reason Group by selecting the Reason Group in the tree widget, making changes and then clicking the Apply button. For an existing Reason Group, these are the fields that can be edited:
 1. Reason Group Description
 2. Parent Reason Group
 - iii. Add Reason Group – Unique Reason Groups are driven by the name of the Reason Group. A new Reason Group can be created by entering a new Reason Group name in the Reason Group text box and clicking the Apply button.
 - iv. Delete Reason Group – to delete a Reason Group, select the record in the tree and click the Delete button
- c. Reasons – the right side of the screen allows for the add, edit and deletion of reasons:

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Edit Reasons

Downtime

Reason	Reason Description	Reason Group	Utilization State	Failure
Unknown	Unknown Downtime	Downtime	Down	☐

Utilization State:

Down

☐ Failure?

Reason:

Unknown

Reason Description:

Unknown Downtime

Apply

Delete

- i. Reasons – Reasons are the granular cause of the current utilization event.
 - ii. Edit Reason – you can edit a Reason by selecting the Reason in the grid, making changes and then clicking the Apply button. For an existing Reason, these are the fields that can be edited:
 1. Utilization State
 2. Failure – flag not currently used by allows for custom or future MTTR and MTBF metrics
 3. Reason Description
 4. Reason Group – will get the currently selected Reason Group in the tree
 - iii. Add Reason – Unique Reasons are driven by the name of the Reason. A new Reason can be created by entering a new Reason name in the Reason text box and clicking the Apply button
 - iv. Delete Reason – to delete a Reason, select the record in the grid and click the Delete button
2. Machine Code – this screen allows a machine to be flagged as a machine that will receive its state data via an integer machine code. The screen also allows for administration of integer machine codes by machine:

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Properties

☒ Use Machine Code to Determine Reason Code

☒ Use Simulated Demo Data

Machine Code(s)

Lathe 1
Delete
Export

Code	Utilization State	Reason Group	Reason
1000	Down	Downtime	Unknown
1001	Down	Electrical	Electric Fault
1002	Down	Mechanical	Conveyor Jam
1003	Down	Mechanical	Feed Issues
1004	Down	Mechanical	Tool Adjustment
1005	Down	Misc	Cleaning
1006	Idle	Idle	No Material

Add or Edit Code

State & Reason: Code:

Downtime
Electrical
Mechanical
Misc
Idle
Running
Setup
Unavailable

Reason	Reason Description	Utilization State	Failure
Unknown	Unknown Downtime	Down	☐

Set Properties
Save

- Flag Machine to get Utilization Event data by machine code – select the machine if interest from the drop-down widget. Flag the “Use Machine Code to Determine Reason Code” check box as true. Click the Set Properties button. This will enable ThingWorx to watch the PTCSCMachineCode integer property on the machine Thing. When that value changes, it will look up from the configured machine codes the corresponding Utilization State, Reason Group and Reason values and then update those properties on the machine Thing.
- Edit Machine Code – you can edit the machine code by selecting the record in the grid, and selecting an existing Reason in the panel below the grid. Once the Reason matching the machine code is selected, click the Save button.
- Add Machine Code – the integer code drives the unique record for a machine. To add a new record, enter a unique integer in the Code numeric entry widget, select a Reason in the lower panel and click the Save button.
- Delete Machine Code – to delete a Machine code record, select the record in the grid and click the delete button to the right of the machine selection drop down widget.

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- e. Note on “Use Simulated Demo Data” check box – you can use services to simulate data for a demo by flagging the “Use Simulated Demo Data” check box as true. This option will not be available if the Utilization configuration for DemoMode is set to false.

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Release Notes

The following table lists application changes starting at version 1.0.3 up to the current version.

Version	Item	Description
1.0.3	Machine Utilization Export Button Not Working	The export button on the Machine Utilization mashup was not connected to any export functionality. A nonvisible data export widget was added and linked to the existing export button.
1.0.3	Mashup Headings to Match Menu	Mashup headings (titles) for the main two mashup were changed to match the menu
1.0.3	Machine Utilization Schedule Chart Heading	Changed the schedule chart heading on the Machine Utilization mashup from “Past hours” to “Utilization Events”
1.0.3	Event Start not Updated When Using Machine Code to Drive Events	This seemed to be functioning correctly in the simulation logic, but added a <code>PTCSC.EventStart = current time</code> to the generic <code>PTCSC.ReasonChange</code> subscription in the <code>PTCSC.Utilization.ModelLogic_TS</code> . Removed any <code>PTCSC.EventStart</code> property setting from the simulation logic. This should ensure that the <code>PTCSC.EventStart</code> is set in a single place – when the <code>PTCSC.Reason</code> property value changes.
1.0.3	Schedule Chart on Machine Utilization Not Updated When Time Fence is Changed	Linked the adjustable time fence timestamps directly to the Scheduler Chart <code>StartTime</code> and <code>EndTime</code> input parameter.
1.0.3	Demo Mode Configuration	Added a <code>DemoMode</code> configuration to each implementation (default = true). If <code>DemoMode</code> is false, the simulated data service will not update any properties and the “Use Simulated Demo Data” check box will not be visible on the Machine Code Admin screen.
1.0.3	Machine Utilization Mashup Drop Down Search	Enabled the list filter functionality on the dropdown widget in the Machine Utilization mashup.
1.0.3	Error Proof Timestamp Selections	Added expressions and event routers to the Utilization Overview and Machine Utilization mashups to error proof time fence selections. If a future end date is manually entered, it will be automatically set to the current time. If a start date is entered that is after the end date, it will be set to the end date. There is some additional logic in the expressions that may need to be tweaked for your specific use cases.
1.0.3	Utilization Overview Machine Click Navigate to Machine Utilization	Enabled navigation in the Utilization Overview mashup. Click on a machine name and navigation to the specific machine in the Machine Utilization mashup will occur.
1.0.3	Toggle Between Utilization % and Hours	Added tabs to Utilization Overview to toggle between State Percentage and State Hours in the Bar Chart. Added tabs to Machine Utilization to toggle between State Percentage and State Hours in the Pie Chart.
1.0.3	Toggle Between Utilization % and Minutes	Added a tab to the State Summary pop-up to toggle between Reason Percentage and Reason Minutes in the Bar Chart.
1.0.4	SCPA Icons	When deployed with SCPA, a unique icon is included on the SCPA menu
1.0.4	SCPA 1.0.4 Add-on	A SCPA specific UI building block that works with this 1.0.4 version of Utilization is now available that uses the SCPA Advanced Search to drill into utilization metrics.
1.0.4	Machine Utilization Mashup Duration Changes	Adjusted the Time in Event metric and the event Duration metric in the grid to be displayed in the h:m:s format.
1.0.4	State Summary Modal Popup Mashup Duration Changes	Adjusted the Average Event Time metric, the Total Time in State metric, and the event Duration in the grid to be displayed in the h:m:s format.
1.0.4	Utilization Event Details Modal Popup Mashup Duration Change	Adjusted the Duration metric to be displayed in the h:m:s format.
1.0.4	Utilization Overview Mashup with Selection	Added a new Utilization Overview Mashup called <code>PTCSC.UtilizationUI.MainWithSelection_UI</code> . This adds some selection capability to the <code>PTCSC.UtilizationUI.Main_MU</code> and replaces the Utilization Overview Mashup in the menu.
1.0.4	Machine Description Services Changes	For non SCPA deployments, the Thing description will be used as the equipment display. If this description is empty or null, the Thing name will be used. If SCPA has been deployed, the equipment display uses the <code>PTCDisplayName</code> (or thing name if not set).
1.0.5	Shift Filtering Capability	Added the ability to filter utilization records by using the existing ShiftID column. There is now a configuration (<code>ShiftFiltersEnabled</code>) that makes the shift filtering screen elements visible and a configuration table (<code>ShiftConfiguration</code>) that allows for the listing of Shifts and Shift Descriptions to be used for filtering.
1.0.5	Reason Edit	By default, only utilization records with the reason “Unknown” can be edited. A configuration was added (<code>ReasonCodeEditEnabled</code>) that allows for any utilization record to be edited.

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1.0.5	Grid Sorting	Column sorting was added to all exposed grid data on the application screens.
1.0.5	Screen Refresh	Added a configuration (InitialScreenRefreshDisabled) allowing for the initial screen refresh to be disabled on application screens.
1.0.5	Schedule Chart Colors	Added a configuration table (UtilizationStateColorConfiguration) where colors can be supplied to the Schedule Charts (Gantt charts) used in the application. The new colors are designed to be more consistent with the other state-based formatting used on other application screen widgets.
1.0.5	Machine Utilization Bug for Machine Properties	Addressed a service bug where, when used with SCPA, some machine properties would not display above the gauge widget on the Machine Utilization screen.
1.0.5	Use Reason Code Change Subscription	Gave the user the ability to not utilize the built in subscription that writes a record to the utilization table when the reason code changes. The subscription now uses a configuration (UseReasonChangeCodeSubscription) to determine if an event record should be written.
1.0.5	Duration Within Time Frame For Specific Machine for Reason, Reason Group or Utilization State	Added generic services to return a duration within a time frame for a specific machine for a given reason, reason group, or utilization state.
1.0.5	Aggregated Duration By Day Within Time Frame For Specific Machine for Reason, Reason Group or Utilization State	Added generic services to return aggregated duration by day within a time frame for a specific machine for a given reason, reason group, or utilization state.
1.0.5	Database Script Changes	Edited the supplied database scripts for both SQL and Postgres to allow the Failure column in the Reason table to be null. Also modified the SQL stored procedure for AddOrUpdateUtilizationDataRecord to use MERGE syntax with some locking parameters.
1.0.5	Reason Edit Filter By State	Added a configuration called FilterReasonCodeEditByState which, when enabled, will only show Reasons when editing a data record that match the current data record's utilization state.
1.0.5	Database thing max row changes	Maximum rows for utilization records to be returned for the GetUtilizationRecordsByMachine service for both the PTCSC.UtilizationPostgresImpl.SQLDatabase_TT and PTCSC.UtilizationSQLImpl.SQLDatabase_TT were set to 10,000,000.

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Enhancement Requests Not Implemented At This Point

The following table lists application requests that have not been implemented at this point.

Item	Category	Description
1	General Enhancement	Use the failure identifier for utilization events to begin to calculate Incident Management Metrics such as MTTR and MTBF. This may involve creating additional data structure as keeping 7 days of data would not allow proper calculation of these metrics. Initial thought would be to save off failure events (with duration).
2	Administration	Make the master data maintenance (State, Reason Group, Reason and Machine Code) a bit more intuitive. Rather than Apply and Delete, move to more of a New, Update and Delete approach.
3	Administration	Ability to Add / Remove States (without having to worry about UI conditional formatting (State Definitions) in the application)

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